Dollar Stablecoins and Arbitrage across Capital Controls

Evidence from the Kimchi Premium

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Master's Thesis

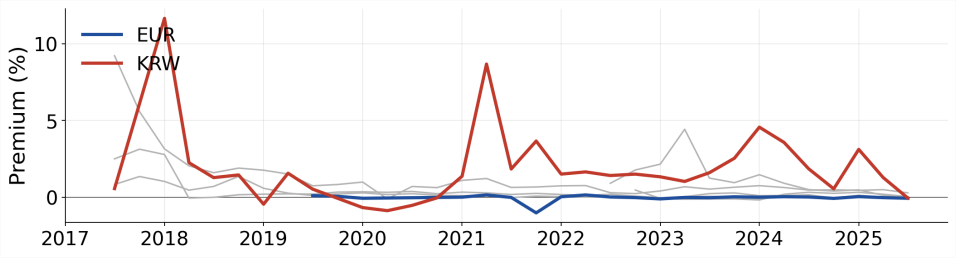
Outline

1. Motivation
2. Literature
3. Research Question and Hypotheses
4. Data
5. Conclusion
6. Appendix

Kimchi premium: paying more for bitcoin?

- The kimchi premium refers to price discrepancies between Korean and offshore exchanges.
- Marginal investors outside the US and Europe are willing to pay more for bitcoin in response to positive news or sentiment.

$$Premium_{Kimchi} = \frac{P_{\text{KR}}^{BTC/KRW}}{FX^{USD/KRW} \cdot P_{\text{GLB}}^{BTC/USDT}} - 1$$

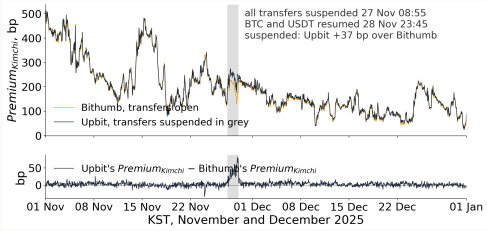


Bitcoin's price on one local exchange per currency over Coinbase's USD price, quarterly mean of daily premia, following Makarov & Schoar (2020). AUD, BRL, GBP, PHP, TRY.

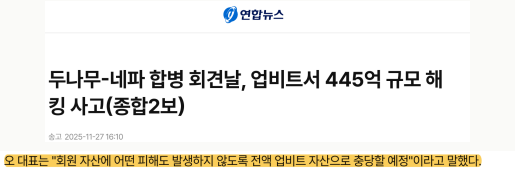
Law of one price fails between Upbit and Bithumb during the suspension

- Cross-exchange arbitrage normally enforces the law of one price.
- It holds even for premiums across countries and exchanges within a country.
- Upbit hack on 27 Nov 2025: suspended **deposits and withdrawals** for 39 hours while Bithumb remained open.
- Under this segmentation, Upbit's $Premium_{Kimchi}$ traded at a persistent premium to Bithumb's.

$$Premium_{Kimchi} = \frac{P_{\mathrm{KR}}^{BTC/KRW}}{FX^{USD/KRW} \cdot P_{\mathrm{GLB}}^{BTC/USDT}} - 1$$



Fence pumping: market manipulation under the suspension


연합뉴스

두나무-네파 합병 회견날, 업비트서 445억 규모 해킹 사고(종합2보)

송고 2025-11-27 16:10

오 대표는 "회원 자산에 어떤 피해도 발생하지 않도록 전액 업비트 자산으로 충당할 예정"이라고 말했다.


TV CHOSUN 뉴스 ● 라이브

경제 > 전체

업비트, 입출금 순차적 재개…'가두리 펌핑'에 수수료 수익은 되레 증가

이정연 기자 | 경제부 등록: 2025.12.01 오후 20:47 • 수정: 2025.12.01 오후 21:21

국내 최대 가상자산 거래소 업비트가 해킹 사고에도 하루 만에 수억 원의 수수료를 벌어들였다.

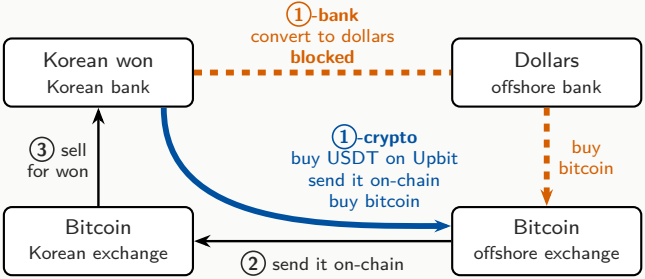
업비트가 지난 27일 445억 원 규모의 해킹 사고 직후 일부 가상화폐 입출금을 중단하면서 이른바 '가두리 펌핑'이 발생해 거래량이 늘어났기 때문이다.

거래소 간 코인 이동이 막히자 업비트 안에서 특정 종목에 매수세가 쏠려 시세가 급등한 것이다.

- *Upbit **temporarily suspended deposit and withdrawal services**; CEO said investors' losses will be covered from Upbit's own assets.*
- *With suspension, buying pressure concentrated on several cryptos and **pushed their prices up**: 'fence pumping'.*

Sources: Yonhap News, 2025-11-27 16:10; TV Chosun, 2025-12-01 20:47. 'Fence pumping': *gaduri pumping*.

Two routes for arbitrage: banking system or blockchain network



- ①-**bank** (fiat-to-crypto): pass through the capital controls on the local fiat currency.
- ①-**crypto** (crypto-to-crypto): bypass offshore banks.

Paying more for bitcoin, or for dollars?

- **USDT** is a cryptocurrency pegged 1-to-1 to the US dollar that moves through blockchain.

$$\underbrace{Premium_{Kimchi}}_{\text{kimchi premium}} \approx \underbrace{Premium_{Banking}}_{\substack{\text{fiat-to-crypto}\\ \text{won-to-dollar fence}}} + \underbrace{Premium_{Crypto}}_{\text{crypto-to-crypto spread}}$$

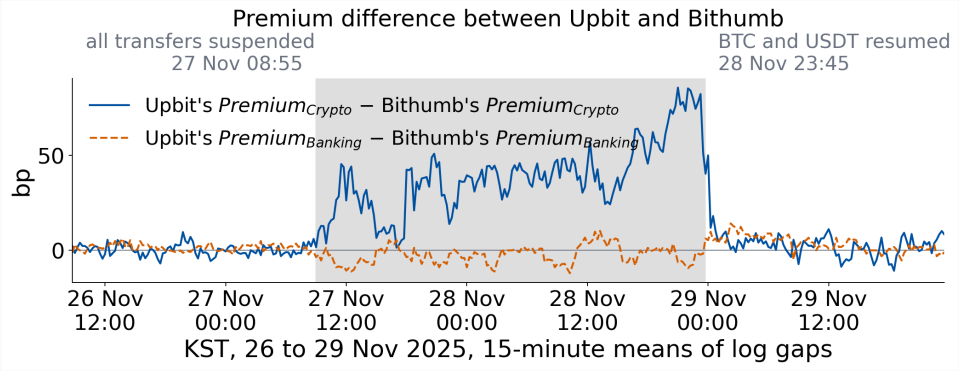
$$\frac{P_{\mathrm{KR}}^{BTC/KRW}}{FX^{USD/KRW} \cdot P_{\mathrm{GLB}}^{BTC/USDT}} - 1 \approx \frac{P_{\mathrm{KR}}^{USDT/KRW}}{FX^{USD/KRW}} - 1 + \frac{P_{\mathrm{KR}}^{BTC/KRW}}{P_{\mathrm{GLB}}^{BTC/USDT} \cdot P_{\mathrm{KR}}^{USDT/KRW}} - 1$$

After the USDT/KRW listing, $Premium_{Banking}$ carries 99% of $Premium_{Kimchi}$'s daily variance on Upbit and Bithumb; $Premium_{Crypto}$ averages under 1 bp.

Makarov & Schoar (2020, *Journal of Financial Economics*)

*"[C]apital controls are most relevant for **the price of crypto to fiat currencies**. In contrast, **transactions between two cryptocurrencies** should not be affected [. . .]."*

Suspend the route, and the pressure stays in $Premium_{Crypto}$



- The suspension blocked the inflow of bitcoin, creating selling pressure on Upbit.
- As a result, $Premium_{Crypto}$ spikes while $Premium_{Banking}$ stays flat.

▸ XRP/KRW in place of USDT/KRW

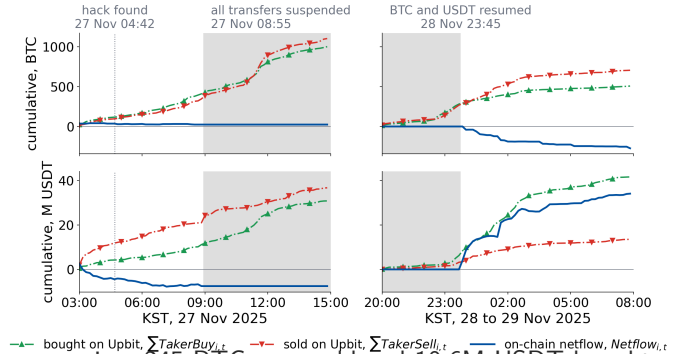
Suspend and resume deposit and withdrawal

The amount of crypto traded at Upbit:

$$NetVol_{i,t} = \sum TakerBuy_{i,t} - \sum TakerSell_{i,t}$$

The amount of crypto transferred on-chain:

$$Netflow_{i,t} = Out_{i,t} - In_{i,t}$$



- In the first six hours of the suspension, 245 BTC were sold and 10.6M USDT bought on Upbit, but none of it could be withdrawn or restocked.
- Once transfers resumed, USDT flowed out at 5.1M per hour, and 188 BTC arrived against 189 sold within the first three hours.

t a ten-minute bin, in crypto i's own units. *TakerBuy*, *TakerSell*: trades a buyer, a seller initiated. *Out*, *In*: i sent to, arriving from Binance, Bybit and OKX on-chain.

Three strands of literature

- **Limits to arbitrage.** Identical assets can trade at different prices across markets because arbitrageurs face capital constraints and risk.
  Shleifer & Vishny (1997, *Journal of Finance*); Froot & Dabora (1999, *Journal of Financial Economics*)

- **The kimchi premium and capital controls.** Crypto premia are largest across borders with capital controls, which bind the fiat-to-crypto price.
  Makarov & Schoar (2020, *Journal of Financial Economics*); Choi, Lehar & Stauffer (2022, working paper, R&R *Journal of Financial and Quantitative Analysis*)

- **Stablecoins in crypto markets.** Stablecoin functions as cross-market numeraire. Griffin & Shams (2020, *Journal of Finance*); Kang, Kang, Kim & Sul (2025, *Economics Letters*)

Contribution

1. Decompose the kimchi premium into fiat-to-crypto and crypto-to-crypto components.
2. Exploit transfer suspensions after exchange hacks as quasi-natural experiments.

Research question and Hypotheses

Research Question

Does a dollar stablecoin carry arbitrage across the capital controls on the local fiat currency, or only across exchanges?

H1 **USDT as arbitraged** USDT/KRW also trades with the kimchi premium, so it serves as a **target** of arbitrage against the foreign exchange rate.

H2 **USDT to arbitrage** USDT serves as the **vehicle** of arbitrage that closes the crypto-to-crypto spread.

Empirical Design: H1

H1. USDT as the arbitraged asset

Daily 2SLS after Upbit's USDT/KRW listing: USDT's round trip against BTC on $Premium_{Kimchi}$, its trading instrumented by its own on-chain netflow.

$$\text{1st stage:} \quad ArbFlow^{NetVol}_{USDT,t} = \pi_0 + \pi_1\, ArbFlow^{\Delta Rsv}_{USDT,t} + \pi' W_{t-1} + \pi_2\, FeeDC_t + u_t$$

$$\text{2nd stage:} \quad \Delta Premium_{Kimchi,t} = \alpha + \beta\, \widehat{ArbFlow^{NetVol}_{USDT,t}} + \gamma' W_{t-1} + \delta\, FeeDC_t + \varepsilon_t$$

$$W_{t-1} = \left(\sigma_{t-1},\ \Delta Premium_{Kimchi,t-1},\ \Delta Premium_{Kimchi,t-1}\,\sigma_{t-1}\right)'$$

- Why instrument: the day's trading can respond to $Premium_{Kimchi}$ as well as move it.
- Instrument: the round trip on-chain, $ArbFlow^{\Delta Rsv}$; first-stage $\chi^2(1)$ and robust endogeneity test reported.
- $Premium_{Banking}$: $\Delta Premium_{Banking,t}$ by OLS on USDT's trading and netflow together, not instrumented (the netflow can move $Premium_{Banking}$ through Upbit's USDT stock).

$ArbFlow^{Flow}_{x,t} = \left(Flow_{x,t} - Flow_{BTC,t}\right)/2$, $Flow \in \{NetVol, \Delta Rsv\}$;
$NetVol_{i,t} = \left(\sum TakerBuy_{i,t} - \sum TakerSell_{i,t}\right) \times P^{i/USDT}_{Binance,t}$; $\Delta Rsv_{i,v,t} = Out_{i,v,t} - In_{i,v,t}$, v = Upbit; both the same day, in dollars. HC1 standard errors.

Empirical Design: H2

H2. USDT as the vehicle currency

USDT flows paired with opposite crypto flows narrow the crypto-to-crypto spread $Premium_{Crypto}$.

- $Premium_{Crypto} > 0$: crypto inflows are sold for KRW and converted to USDT, which flows out, narrowing it.
- $Premium_{Crypto} < 0$: USDT inflows are converted to KRW to buy crypto, which flows out, narrowing it.

Event study of the 2025 Upbit hack, one-hour bins on one-minute books, BTC and ETH pooled:

$$10^4 \ln\left(\frac{P^i_{Upbit,m}}{P^i_{Bithumb,m}}\right) = \mu_i + \sum_{k \neq -1} \theta_k \mathbb{1}[\text{hour}(m) = k] + \varepsilon_{i,m}$$

Data

12 crypto assets (BTC, ETH, XRP, USDT, . . .) on **Upbit** and **Bithumb**, against **Binance**.

Data	Frequency	Provider	Coverage	Period
Prices	Hourly, minute	Upbit, Bithumb, Binance, Bybit, OKX	12 assets, KRW and USDT	2017-10-24 – 2026-09-25
FX rate	Minute-level	Barchart	USD/KRW interbank	2017-10-24 – 2026-08-28
Trades	Tick-level	Upbit	17 books (BTC, USDT, . . .)	2022-05-01 – 2026-09-25
Holdings	Daily	Arkham Intelligence, XRP Ledger, Hyperion	12 assets, both venues	2017-10-24 – 2026-09-24
Transfers	Transaction-level	Arkham Intelligence, XRP Ledger	6 assets, 2.89M transfers	2019-02-28 – 2026-02-28
Notices		Upbit, Bithumb	501 assets, 2,791 notices	2017-12-12 – 2026-09-24

On file within 2017-10-24 – 2026-09-25, UTC. Prices: hourly from 2017-10-24; minute books from 2019-01-01 (Upbit, Bithumb), 2017-10-24 (Binance). Hyperion: EOS from 2023-03.

Conclusion and policy implications

Conclusion

1. The kimchi premium is the price of the won-to-dollar fence, not of bitcoin: a dollar stablecoin does not relax the frictions on the local fiat currency.
2. A dollar stablecoin serves as the vehicle of arbitrage that moves pressure from $Premium_{Crypto}$ into $Premium_{Banking}$; when transfers are suspended, the pressure stays in $Premium_{Crypto}$.

Policy implications

1. Listing a dollar stablecoin against the won does not integrate the won with the dollar.
2. Capital controls force arbitrageurs to hold bitcoin in Korea in advance, exposing them to price risk.

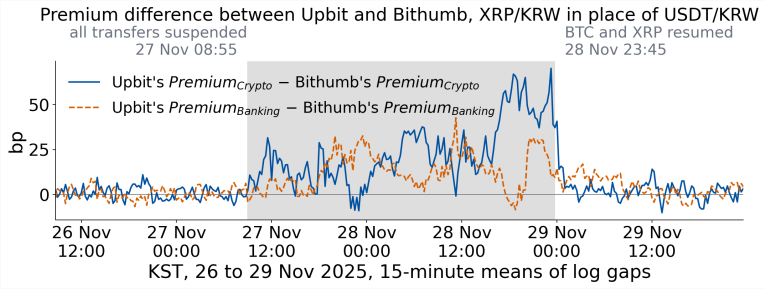
Appendix

Korea won's capital controls

Rule	In force	Source
Remittance cap	A resident may send abroad without documents at most **USD 50,000** a year (1999 to 3 July 2023), **USD 100,000** a year since 4 July 2023, and **USD 5,000** per transfer (from Jan 2026)	Ministry of Economy and Finance (10 Feb 2023; 8 Dec 2025)
Exchange access	Won deposits require a real-name bank account; **non-residents cannot trade** on Korean exchanges (since 30 Jan 2018)	Financial Services Commission (23 Jan 2018)
Enforcement	**USD 12.26 billion** of outward remittances by 84 firms, collected from crypto exchanges and sent abroad as import payments, 2021–22	Financial Supervisory Service (4 Apr 2023)

XRP/KRW in place of USDT/KRW

$$Premium_{Banking} = \frac{P_{\mathrm{KR}}^{XRP/KRW}}{FX^{USD/KRW}} - 1 \qquad Premium_{Crypto} = \frac{P_{\mathrm{KR}}^{BTC/KRW}}{P_{\mathrm{GLB}}^{BTC/USDT} \cdot P_{\mathrm{KR}}^{XRP/KRW}} - 1$$



- While Upbit's transfers are suspended, $Premium_{Crypto}$'s gap averages **+23 bp** and $Premium_{Banking}$'s **+13 bp**: XRP's won price is pumped too.
- With USDT/KRW, $Premium_{Banking}$ stays level: a dollar stablecoin's won price holds when its transfers are cut off.

◂ Which route closed, which spread diverged?

Variables

Variable	Definition	Tables
$Premium_{Kimchi,i,v,t}$	$P_{v,t}^{i/KRW} \,/\, \left(FX_t^{USD/KRW} \cdot P_{GLB,t}^{i/USDT}\right) - 1$	3, 5, 6, 11, 12
$Premium_{Banking,v,t}$	$P_{v,t}^{USDT/KRW} \,/\, FX_t^{USD/KRW} - 1$	3, 5
$Premium_{Crypto,x,v,t}$	$P_{v,t}^{BTC/KRW} \,/\, \left(P_{GLB,t}^{BTC/USDT} \cdot P_{v,t}^{x/KRW} / P_{GLB,t}^{x/USDT}\right) - 1$	3, 9, 10, Fig. 4
$\frac{Premium_{Kimchi,i}}{Premium_{Kimchi,-i}}$	$Premium_{Kimchi,i,v,t} - \frac{1}{N-1}\sum_{j \neq i} Premium_{Kimchi,j,v,t}$	8
$\Delta Rsv_{i,v,t}$	$\left(Rsv_{i,v,t} - Rsv_{i,v,t+1}\right) \times P_{Binance,t}^{i/USDT} = Out_{i,v,t} - In_{i,v,t}$	5, 6, 7, 8
$Netflow_{i,t}$	$Out_{i,t} - In_{i,t}$ s.t. t: ten-minute bin	9
$NetVol_{i,t}$	$\left(\sum TakerBuy_{i,t} - \sum TakerSell_{i,t}\right) \times P_{Binance,t}^{i/USDT}$	5, 6
$NetTurnover_{i,t}$	$\left(\sum TakerBuy_{i,t}^{KRW} - \sum TakerSell_{i,t}^{KRW}\right) / FX_t^{USD/KRW}$	9, 10, Fig. 4
$ArbFlow_{x,t}^{Flow}$	$\left(Flow_{x,t} - Flow_{base,t}\right)/2$ s.t. $Flow \in \{NetVol, \Delta Rsv\}$, $base \in \{BTC, USDT\}$	5, 6
$Treat \times Post$	$\mathbf{1}[v = v^*] \cdot \mathbf{1}[h \geq h^*]$ s.t. v^*: hacked exchange, h^*: event hour	11, 12, A8
I	$\max_t \left(G_t - \min_{s \leq t} G_s\right), \quad G_t = \sum_{u \leq t} \left(Sold_u - Restock_u\right)$	Fig. 4

The thesis's tables write K, B and C; the slides write $Premium_{Kimchi}$, $Premium_{Banking}$ and $Premium_{Crypto}$.